METHODS OF EXPLORATION

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Problem 1. Is there an isosceles triangle that can be cut into 3 triangles such that we can put any two of them together to form an isosceles triangle?

Problem 2. Prove that there are infinitely many positive integers n such that it is possible to place 1 to $3n$ into n groups (each group has 3 numbers) and the sum of the two smaller numbers in each group equals the third.

Problem 3. n assistants start simultaneously from one vertex of a cube-shaped planet with edge length 1. Each assistant moves along the edges of the cube at a constant speed of $2, 4, 8, \cdots, 2^n$, and can only change their direction at the vertices of the cube. The assistants can pass through each other at the vertices, but if they collide at any point that is not a vertex, they will explode. Determine the maximum possible value of n such that the assistants can move infinitely without any collisions.

Problem 4. For positive integers a and b, write $a \smile b$ if $ab+1$ is a perfect square. Prove that if $a \smile b$, then there exists a positive integer c such that $a \smile c$ and $b \smile c$.

Problem 5. A rectangle R is divided into a set S of finitely many smaller rectangles with sides parallel to the sides of R such that no three rectangles in S share a common corner. An ant is initially located at the bottom-left corner of R. In one operation, we can choose a rectangle $r \in S$ such that the ant is currently located at one of the corners of r, say c, and move the ant to one of the two corners of r adjacent to c. Suppose that after a finite number of operations, the ant ends up at the top-right corner of R. Prove that some rectangle $r \in S$ was chosen in at least two operations.

Problem 6. A positive integer k is given. Initially, N cells are marked on an infinite checkered plane. We say that the cross of a cell A is the set of all cells lying in the same row or in the same column as A. By a turn, it is allowed to mark an unmarked cell A if the cross of A contains at least k marked cells. It appears that every cell can be marked in a sequence of such turns. Determine the smallest possible value of N.

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Problem 7. Let $n > 1$ be an integer. In a configuration of an $n \times n$ board, each of the n^2 cells contains an arrow, either pointing up, down, left, or right. Given a starting configuration, Turbo the snail starts in one of the cells of the board and travels from cell to cell. In each move, Turbo moves one square unit in the direction indicated by the arrow in her cell (possibly leaving the board). After each move, the arrows in all of the cells rotate $90°$ counterclockwise. We call a cell good if, starting from that cell, Turbo visits each cell of the board exactly once, without leaving the board, and returns to her initial cell at the end. Determine, in terms of n, the maximum number of good cells over all possible starting configurations.

Problem 8. Given positive integer $n \geq 2$. Now Cindy fills each cell of the $n \times n$ grid with a positive integer not greater than n such that the numbers in each row are in a non-decreasing order (from left to right) and numbers in each column is also in a non-decreasing order (from top to bottom). We call two adjacent cells form a *domino* , if they are filled with the same number. Now Cindy wants the number of *domino*s as small as possible. Find the smallest number of *dominos* Cindy can reach. (Here, two cells are called adjacent if they share one common side)